

Gurpreet Singh Deol

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Professional summary

Data scientist with an MSc in Data Science and a background in Physics, specialising in machine learning, geospatial analytics, and data pipeline development. Experienced in building end-to-end data solutions using Python, PostgreSQL/PostGIS, and cloud tools as well as working with large real-world datasets, developing ML models, and deploying interactive analytical applications with Streamlit.

Education

Royal Holloway University of London - Egham, Surrey

MSc Data Science and Analytics: Pass with Merit

Oct 2021 - Oct 2022

Key modules: Data Analysis, Machine Learning, Artificial Intelligence, Databases and Business Intelligence.

- First-class dissertation: Applied PCA and K-Means clustering to analyse tactical evolution in football across 13 seasons, identifying a league-wide shift from direct play to possession-based strategies.
- Completed multiple projects applying supervised and unsupervised machine learning techniques to real-world datasets.

BSc Physics: Second Class Honours (Upper Division), 2:1

Sep 2018 - Jul 2021

Key modules: Mathematical Methods, Statistics, Quantum Mechanics, Particle Physics and Energy and Climate Science.

- First-class dissertation: Simulated electrical resistance, current and potential fields in 2D geometries using Python.
- Developed strong analytical and data handling skills by working with datasets ranging from astronomical observations to experimental physics measurements, producing clear and comprehensive analytical reports.

IBM Data Science Professional Certificate

March 2026

- Completed modules in machine learning, Python, SQL and dashboard development, with practical project work in data analysis and visualisation.

Skills

- Programming Languages: Python, SQL, R, MATLAB
- Data Science & Machine Learning: Pandas, NumPy, Scikit-Learn, TensorFlow, Keras, PySpark
- Data Engineering: PostgreSQL, PostGIS, Docker, ETL pipelines, API integration
- Cloud: AWS (Lambda, EC2, S3), Azure
- Visualisation: Power BI, Streamlit, Matplotlib, Seaborn
- Tools: Git, GitHub, Jupyter, VS Code, LaTeX

Projects

Maritime IUU Fishing Detection with LightGBM | [GitHub](#) | [Web Application](#)

- Built a LightGBM classifier to detect fishing activity from raw AIS vessel tracking data, achieving 94% accuracy.
- Fetched over 1M positional pings directly from the Global Fishing Watch API and constructed rolling 2-hour windows across 10,000+ vessel tracks.
- Engineered 15+ physics-based features including speed variance, turning rate and course deviation from raw AIS pings.
- Scored vessel activity against Marine Protected Area boundaries across 3 protection categories to flag potential IUU fishing.
- Deployed a Streamlit web application for risk score visualisation, vessel track mapping and feature inspection.

Public Health and Air Quality Dashboard | [GitHub](#) | [Live Dashboard](#)

- Built an end-to-end ETL pipeline collecting pollution, weather, and health data for 10 global cities using Python APIs.
- Automated ingestion and storage using Docker + PostgreSQL.
- Processed WHO datasets to analyse relationships between air pollution and mortality rates.
- Developed an interactive Streamlit dashboard with geospatial visualisations and time-series analytics.
- Created Power BI dashboards for long-term trend analysis.

Sports Data Analysis: Football and Machine Learning | [GitHub](#)

- Built a dataset of 2,000+ team-season records with 100+ performance features from historical football data.
- Performed feature engineering using Pandas to construct tactical and performance metrics.
- Applied Principal Component Analysis and K-Means clustering to identify long-term tactical shifts in football.
- Visualised patterns using Matplotlib, presenting results in a technical report written in LaTeX.

Work experience

Computer Science Teacher - North Brent School – Wembley Multi Academy Trust - London, NW10 2UF

Sep 2025 - Present

- Teach Computer Science to 400+ students across multiple year groups at one of the highest-performing academy trusts in the UK.
- Plan and deliver lessons covering Python programming, data structures, and computational thinking, adapting teaching approaches to support students with different ability levels.
- Design learning materials and programming exercises aimed at improving problem-solving and exam performance.
- Analyse student assessment data using Python and Excel to identify performance trends and adjust teaching strategies to better support student progress.